Exercise sheet 1: Introduction to stochastic calculus

The notations are the same as in the course. In all exercises, we consider a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{F})$ as well as an $\mathbb{F}$-Brownian motion $(B_t)_{t\geq 0}$.

Exercice 1. Soit $t \geq 0$.

1. Determine the characteristic function of B_t, denoted by ϕ_{B_t}.

2. Calculate $\mathbb{E}\left[|B_t|\right]$ and $\mathbb{E}\left[B_t \mathbb{1}_{\{B_t \leq a\}}\right]$ for all $a \in \mathbb{R}$.

3. Let X be a random variable with distribution $\mathcal{N}(0,1)$. Calculate $\mathbb{E}[X^4]$. Deduce the value of $\mathbb{E}[B_t^4]$.

Exercice 2. Let X be a stochastic process such that for all $0 \leq s < t$, $X_t - X_s$ follows a $\mathcal{N}(0, t-s)$ distribution. Show that

$$\mathbb{E}\left[|X_t - X_s|^4\right] \leq 3|t-s|^2.$$

Exercice 3. Let $\mu \in \mathbb{R}$ and $\sigma, S_0 > 0$. We define the process $(S_t)_{t\geq 0}$ by

$$S_t = S_0 e^{\mu t + \sigma B_t}.$$

1. Let X be a random variable with distribution $\mathcal{N}(0,1)$. Show that for all $u \in \mathbb{R}$,

$$\mathbb{E}[e^{uX}] = e^{u^2/2}.$$

Deduce the expectation and variance of S_t for all $t \geq 0$.

2. Let $\phi : \mathbb{R} \to \mathbb{R}$ be a measurable and bounded function. Show that for all $0 \leq s \leq t$,

$$\mathbb{E}\left[\phi(S_t)|\mathcal{F}_s\right] = \varphi(S_s), \quad \text{where } \varphi : x \in \mathbb{R} \mapsto \mathbb{E}\left[\phi\left(x\frac{S_t}{S_s}\right)\right].$$

Exercice 4. Show that each of the following processes is an $\mathbb{F}$-martingale.

1. $(B_t)_{t\geq 0}$.

2. $\left(B_t^2 - t\right)_{t\geq 0}$.

3. $\left(B_t^3 - 3tB_t\right)_{t\geq 0}$.

4. $\left(e^{-\frac{\sigma^2}{2}t + \sigma B_t}\right)_{t\geq 0}$, with $\sigma > 0$.

Exercice 5. Correct the formulas below so that the following statements are true.

1. $(2 + B_t)_{t\geq 0}$ is an $\mathbb{F}$-martingale.

2. $(2 + B_{2t})_{t\geq 0}$ is an $\mathbb{F}$-martingale.

3. $\left(\frac{B_{3t}}{\sqrt{2}}\right)_{t\geq 0}$ is a Brownian motion.

4. $\left(\frac{tB_{4/t}}{\sqrt{2}}\right)_{t\geq 0}$ is a Brownian motion.

Exercice 6. Show that the Brownian motion B is a Gaussian process. Determine $e_B(t) := \mathbb{E}[B_t]$ and $K_B(s,t) := Cov(B_t, B_s)$ for all $0 \leq s \leq t$.

Exercice 7. Let $(M_t)_{t\geq 0}$ be a continuous martingale satisfying $M_0 = 0$ such that $\left(M_t^2 - t\right)_{t\geq 0}$ is also a martingale. Suppose further that M is a Gaussian process. Show that M is a Brownian motion.

Exercice 8. Show that the process $(X_t)_{t\geq 0}$ is an $\mathbb{F}$-standard Brownian motion if and only if for all $\lambda \in \mathbb{R}$, the complex process $\left(M_t^\lambda\right)_{t\geq 0}$ defined by

$$M_t^\lambda = e^{i\lambda X_t + \frac{\lambda^2}{2}t} \quad \forall t \geq 0$$

is a continuous $\mathbb{F}$-martingale such that $M_0^\lambda = 1$.

Exercice 9. Let $x > 0$ and $A = \left\{ \overline{\lim_{n\to+\infty}} \frac{B_n}{\sqrt{n}} \geq x \right\}$.

1. Show that for all $m \in \mathbb{N}$, the event A is independent of $\mathcal{F}_m$. Conclude that $\mathbb{P}(A) \in \{0,1\}$.

2. Show that $\mathbb{P}(A) = 1$. Conclude that

$$\overline{\lim_{n\to+\infty}} \frac{B_n}{\sqrt{n}} = +\infty \quad \text{and} \quad \underline{\lim_{n\to+\infty}} \frac{B_n}{\sqrt{n}} = -\infty \quad \text{a.s.}$$

3. Conclude that for all $t \geq 0$ and for all $x \in \mathbb{R}$,

$$\mathbb{P}\left(\exists s \geq t, B_s = x\right) = 1.$$

Exercice 10. Let $t \geq 0$. Using the properties of translation invariance and time reversal, show that almost surely the Brownian motion is not differentiable at t.

Exercice 11. Let a process $Z = (Z_t)_{t\in[0,1]}$ be such that $\forall t \in [0,1]$,

$$Z_t = B_t - tB_1.$$

1. Show that Z is a Gaussian process independent of B_1.

2. Determine the functions $e_Z(t) := \mathbb{E}[Z_t]$ and $K_Z(s,t) := Cov(Z_s, Z_t)$.

3. Show that the process $\tilde{Z}$ defined by $\tilde{Z}_t = Z_{1-t}$ has the same distribution as Z.

Exercice 12. Let $\Delta = \{0 = t_0 < t_1 < \ldots < t_N = T\}$ be a subdivision of the interval $[0,T]$.

1. Calculate the L^2 limit of the Riemann sum

$$S_1 := \sum_{k=0}^{N-1} B_{t_k}\left(B_{t_{k+1}} - B_{t_k}\right)$$

as $|\Delta| \to 0$.

2. Calculate the L^2 limit of the Stratonovich sum

$$S_2 := \sum_{k=0}^{N-1} B_{\frac{t_k+t_{k+1}}{2}}\left(B_{t_{k+1}} - B_{t_k}\right)$$

as $|\Delta| \to 0$.

Exercice 13. Consider a continuous function $f : \mathbb{R} \to \mathbb{R}$. For all $t \geq 0$, we define the random variable

$$X_t^f := \int_0^t f(s) B_s \mathrm{d}s.$$

1. Show that for all $t \geq 0$, X_t^f is well-defined and

$$X_t^f = \lim_{n \to +\infty} X_t^{f,n}$$

almost surely, where

$$X_t^{f,n} = \frac{t}{n} \sum_{k=0}^{N-1} B_{\frac{k}{n}t} f\left(\frac{k}{n} t\right).$$

2. Show that for all $t \geq 0$ and $n \geq 1$, $X_t^{f,n}$ is a Gaussian variable. Calculate its mean and variance. Deduce the distribution of x_t^f.

3. Calculate $\mathbb{E}\left[\int_0^t B_s \mathrm{d}s\right]$ and $\mathbb{E}\left[\left(\int_0^t B_s \mathrm{d}s\right)^2\right]$.

4. Is the process $M = (M_t)_{t \geq 0} = \left(\int_0^t B_s \mathrm{d}s\right)_{t \geq 0}$ a Brownian motion ? A martingale ?

Exercice 14. Let $0 < T < +\infty$ and $\lambda > 0$. We consider the process $(L_t)_{t \in [0,T]}$ given by

$$L_t := e^{\lambda B_t - \frac{\lambda^2}{2} t}.$$

1. Justify that we can define on $(\Omega, \mathcal{F}_T)$ a probability measure $\mathbb{Q}_T$ equivalent to $\mathbb{P}$ by setting $\frac{\mathrm{d}\mathbb{Q}_T}{\mathrm{d}\mathbb{P}} = L_T$.

2. Let $t \in [0, T]$ and $\mathbb{Q}_t$ the probability measure with density L_t with respect to $\mathbb{P}$. Show that $\mathbb{Q}_t$ and $\mathbb{Q}_T$ coincide on $\mathcal{F}_t$.

3. Let Z be a bounded and $\mathcal{F}_T$-measurable random variable. Show that

$$\mathbb{E}^{\mathbb{Q}_T}[Z|\mathcal{F}_t] = \frac{\mathbb{E}^{\mathbb{P}}[Z L_T|\mathcal{F}_t]}{L_t}.$$

4. Show that the process $\left(B_t^\lambda\right)_{t \in [0,T]}$ defined by

$$B_t^\lambda = B_t - \lambda t$$

is a Brownian motion on $(\Omega, \mathcal{F}_T, \mathbb{Q}_T)$.

5. Explain why it is not possible to take $T = +\infty$.

Exercice 15. Let $t > 0$ fixed. For all $n \geq 1$, we denote S^n the random variable

$$S^n = \sum_{k=1}^{2^n} \left(B_{\frac{k}{2^n}t} - B_{\frac{k-1}{2^n}t}\right)^2.$$

1. Calculate $\mathbb{E}[S^n]$ and $\mathbb{V}(S^n)$.

2. Using the Bienaymé-Tchebychev inequality, show that for all $\varepsilon > 0$,

$$\mathbb{P}(|S^n - t| > \varepsilon) \leq \frac{4t^2}{\varepsilon^3} 2^{-2n}.$$

3. Deduce that S^n converges to t almost surely.

Exercice 16. Let $(M_t)_{t \geq 0}$ be a martingale with square integrable variations.

1. Verify that for all $0 \leq s \leq t$, $\mathbb{E}\left[(M_t - M_s)^2\right] = \mathbb{E}\left[(M_t)^2\right] - \mathbb{E}\left[(M_s)^2\right]$.

2. Suppose that M has continuous paths. Suppose furthermore that the variations of M are bounded : there exists a constant $C > 0$ such that for all $t > 0$,

$$\mathbb{P}\left(V_t^{(1)}(M) \leq C\right) = 1.$$

a. Verify that for any subdivision $\Delta = \{0 = t_0 < t_1 < \ldots < t_N = t\}$ of the interval $[0, t]$, we have

$$\mathbb{E}\left[(M_t - M_0)^2\right] \leq \mathbb{E}\left[\left\{\sup_i |M_{t_i} - M_{t_{i-1}}|\right\} V_t^{(1)}(M, \Delta)\right].$$

b. Deduce that $M = M_0$ almost surely.

Exercice 17. Let $T > 0$ be fixed. The goal of this exercise is to study the stochastic integral of the form

$$\int_0^t f(s)\mathrm{d}Bs$$

for $t \in [0, T]$ and $f \in L^2([0, T])$. Let $\mathcal{S}([0, T])$ be the set of functions f of the form

$$f : s \mapsto \sum_{k=0}^{N-1} a_k \mathbb{1}_{]t_k, t_{k+1}]}(s)$$

for a subdivision $\Delta = \{0 = t_0 < t_1 < \ldots < t_N = T\}$ of $[0, T]$ and $(a_k)_{0 \leq k \leq N-1} \in \mathbb{R}^N$.

1. For a function $f = \sum_{k=0}^{N-1} a_k \mathbb{1}_{]t_k, t_{k+1}]}$ in $\mathcal{S}([0, T])$, we define

$$I_{\mathcal{S}}(f) := \sum_{k=0}^{N-1} a_k \left(B_{t_{k+1}} - B_{t_k}\right).$$

Show that $I_{\mathcal{S}}(f)$ is a Gaussian random variable. Determine its mean and variance.

2. It is admitted that the space $\mathcal{S}([0, T])$ is dense in $L^2([0, T])$. Show that there exists a unique linear map I defined on $L^2([0, T])$ with values in $L^2(\Omega)$ such that for all $f \in \mathcal{S}([0, T])$, $I(f) = I_{\mathcal{S}}(f)$, and for all $f \in L^2([0, T])$,

$$\mathbb{E}\left[I(f)^2\right] = ||f||_{L^2}^2 = \int_0^T f(s)^2 \mathrm{d}s.$$

3. Let $f \in L^2([0, T])$. We define the process $(M_t(f))_{t \in [0, T]}$ by

$$M_t(f) = I\left(f \mathbb{1}_{[0, t]}\right).$$

For all $t \in [0, T]$, what is the distribution of $M_t(f)$? Show that $M(f)$ is an $\mathbb{F}$-adapted process and that for all $0 \leq s \leq t \leq T$, $M_t(f) - M_s(f)$ is a random variable independent of $\mathcal{F}_s$.

4. Let $f \in L^2([0, T])$. We consider the process $(L_t)_{t \in [0, T]}$ given by

$$L_t = e^{\int_0^t f(s)\mathrm{d}Bs - \frac{1}{2}\int_0^t f(s)^2 \mathrm{d}s}.$$

a. Show that L is a martingale.

b. Let $\mathbb{Q}_T$ be the probability measure on Ω with density L_T with respect to $\mathbb{P}$. Show that the process $\left(B_t^f\right)_{t \in [0, T]}$ given by

$$B_t^f = B_t - \int_0^t f(s)\mathrm{d}s$$

is a Brownian motion on $(\Omega, \mathcal{F}_T, \mathbb{Q}_T)$.

Exercice 18. Let B be a standard Brownian motion.

1. Show that for all $t \geq 0$, the variable
$$X_t = \int_0^t \sin(s)\mathrm{d}B_s$$
is well-defined.

2. Show that $X = (X_t)_{t\geq 0}$ is a Gaussian process. Give its distribution.

3. Show that for all $t \geq 0$, we have
$$X_t = \sin(t)B_t - \int_0^t \cos(s)B_s\mathrm{d}s.$$

4. Show that the process $Y = (Y_t)_{t\geq 0}$ given by
$$Y_t = \sin(B_t) + \frac{1}{2}\int_0^t \sin(B_s)\mathrm{d}s$$
is a martingale.

Exercice 19. Let $T, \sigma > 0$ be fixed.

1. Show that the process $M = (M_t)_{t\in[0,T]}$, given by
$$M_t = \int_0^t e^{\sigma B_s}\mathrm{d}B_s$$
is well-defined. Show that M is a martingale, and determine $\mathbb{E}[M_t]$ and $\mathbb{V}(M_t)$.

2. Same questions for $N = (N_t)_{t\in[0,T]}$ given by
$$N_t = \int_0^t M_s\mathrm{d}B_s.$$

Exercice 20. We define the random variables
$$X = \int_0^1 \int_0^t 1\mathrm{d}s\mathrm{d}B_t$$
and
$$X = \int_0^1 \int_0^t 1\mathrm{d}Bs\mathrm{d}t.$$

1. Calculate with, then without using the Itô formula, the covariance of the random variables X and Y.

2. Are the two random variables equal ?

Exercice 21. Apply Itô's formula to the following processes :

1. $X_t = B_t^3, \quad t \geq 0.$

2. $C_t = e^{\frac{t}{2}}\cos(B_t), \quad t \geq 0.$

3. $D_t = e^{\frac{t}{2}}\sin(B_t), \quad t \geq 0.$

4. $K_t = (B_t + t)e^{-B_t - \frac{t}{2}}, \quad t \geq 0.$

5. $S_t = S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma B_t}, \quad t \geq 0$, with $S_0, \sigma > 0$ and $\mu \in \mathbb{R}$.

6. $Y_t = u(t, S_t), \quad t \geq 0$, where u is a $\mathcal{C}^{1,2}$ function.

7. $V_t = \sqrt{S_t}, \quad t \geq 0.$

8. $Z_t = e^{rt}(S_t)^2, \quad t \geq 0$, where $r \in \mathbb{R}$.

9. $R_t = e^{-\int_0^t r_s ds} N_t, \quad t \geq 0$, with

$$r_t = r_0 + \int_0^t \alpha(\beta - r_s)\mathrm{d}s + \int_0^t \sigma \mathrm{d}B_s,$$

$$N_t = N_0 + \int_0^t r_s N_s \mathrm{d}_s + \int_0^t \sigma N_s \mathrm{d}B_s,$$

where $\alpha, \beta, \sigma, r_0$ and N_0 are real constants.

Exercice 22. Let $T > 0$. Find the process $(H_t^T)_{t\in[0,T]}$ such that

$$B_T^3 = \int_0^T H_t^T \mathrm{d}B_t.$$

Start by writing $Y_T = \int_0^T B_t \mathrm{d}t$ in a similar form.

Exercice 23. Let $f : \mathbb{R} \to R$ be a continuous and bounded function. Let $u : [0,T] \times \mathbb{R} \to \mathbb{R}$ be a $\mathcal{C}^{1,2}$ function that satisfies

$$\begin{cases} \partial_t u(t,x) = \frac{1}{2}\partial_{xx}^2 u(t,x) & \forall (t,x) \in]0,T] \times \mathbb{R}, \\ u(0,x) = f(x) & \forall x \in \mathbb{R}. \end{cases}$$

Assume moreover that $\partial_x u$ is bounded. Show then that

$$u(t,x) = \mathbb{E}[f(x + B_t)].$$

Exercice 24. *1.* Let $\mu, \sigma \in \mathbb{R}$. Show that the following SDE has a unique solution :

$$\begin{cases} \mathrm{d}S_t = \mu S_t \mathrm{d}t + \sigma S_t \mathrm{d}B_t, \\ S_0 \in \mathbb{R} \text{ fixed.} \end{cases}$$

Show that this solution is given by

$$S_t = S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma B_t}.$$

2. Let $\alpha \in \mathbb{R}^*$, $\beta \in \mathbb{R}$, and $\sigma > 0$. Show that the following SDE has a unique solution :

$$\begin{cases} \mathrm{d}r_t = \alpha(\beta - r_t)\mathrm{d}t + \sigma \mathrm{d}B_t, \\ r_0 \in \mathbb{R} \text{ fixed.} \end{cases}$$

Show that this solution is given by

$$r_t = r_0 e^{-\alpha t} + \beta(1 - e^{-\alpha t}) + \sigma e^{-\alpha t} \int_0^t e^{\alpha s} \mathrm{d}B_s.$$

3. Let $\alpha \in \mathbb{R}^*$, $\beta \in \mathbb{R}$, and $\sigma > 0$. Does the result from the course allow us to claim that the following SDE has a unique solution ?

$$\begin{cases} \mathrm{d}r_t = \alpha(\beta - r_t)\mathrm{d}t + \sigma\sqrt{|r_t|}\mathrm{d}B_t, \\ r_0 \geq 0 \text{ fixed.} \end{cases}$$